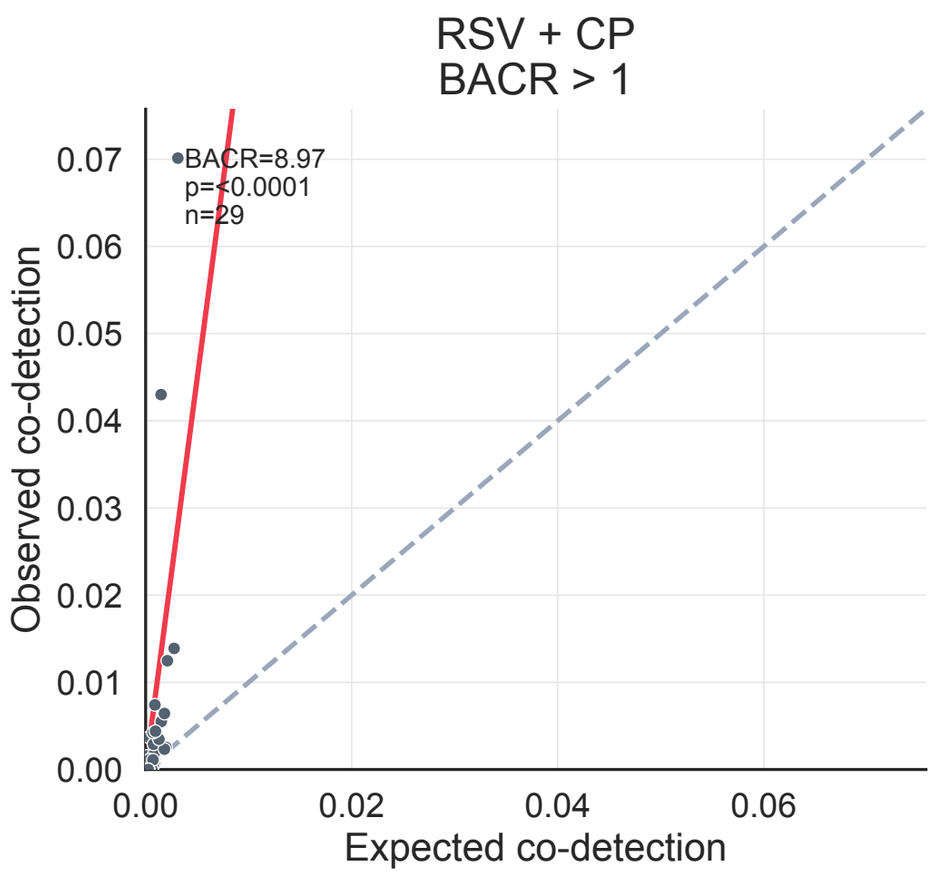
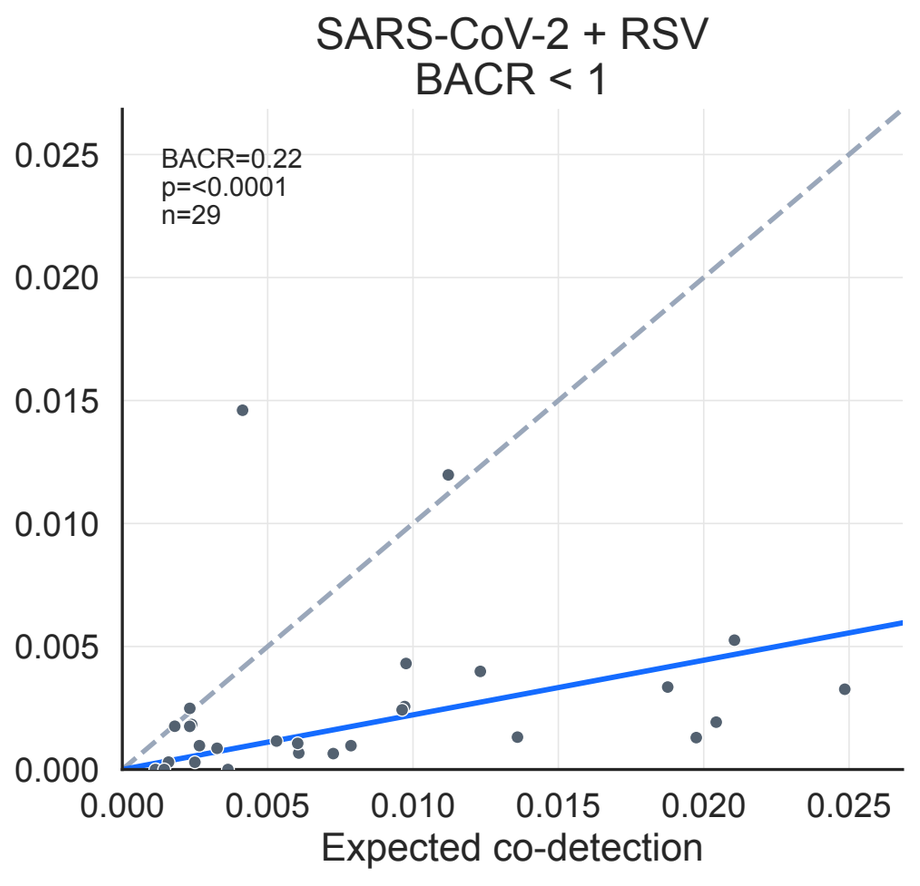
Monthly positivity
$P_t(a)$ $P_t(b)$

↓

Expected co-detection
$P_t(a) \times P_t(b)$

↓

Observed co-detection
$O_t(ab)$

↓

BACR-adjusted estimate
$P_t(a) \times P_t(b) \times BACR_{a,b}$